

Effects of constant and sinusoidal display lag on sickness during active exposures to virtual reality

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ABSTRACT

When we move our heads in virtual reality (VR), display lag creates differences between our virtual and physical head pose (DVP). This study examined whether objective estimates of these DVP could be used to predict the sickness caused by different types of lag. We found that adding constant and time-varying lag to simulations generated similar levels of sickness – with all added lag conditions producing more severe sickness than our baseline control. Consistent with the *DVP hypothesis*, the spatial magnitude and temporal dynamics of the DVP were both found to predict cybersickness severity during active HMD VR.

Keywords: Head-mounted display, Cybersickness, Motion-to-photon latency, Sensory conflict.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Virtual Reality—Empirical studies in HCI

1 INTRODUCTION

Sensory conflicts due to display lag (or motion-to-photon latency) are known to induce cybersickness during HMD VR. A number of studies have examined the effects of adding extra lag into their VR simulations (see [1] for a review). Most of them injected constant lags on top of the system's baseline lag. Sickness was generally found to increase with the magnitude of this constant lag. However, adding constant lag did not always exacerbate cybersickness (e.g., [2]). These mixed findings could have occurred because: 1) the systems used already had quite high baseline lags (40 ms or greater) – making it harder to detect the effects of adding extra lag; and 2) the physical movement tasks examined were quite different across these past lag studies. However, St. Pierre et al. [2] argued that constant lags are not typical, or provocative for sickness, during HMD VR. Consistent with their proposal, they found that sickness was more severe when sinusoidal time-varying lag was added to their simulations (compared to baseline and constant lag conditions).

According to the *DVP hypothesis* [1, 3, 4], cybersickness cannot be understood based on display lag alone – knowledge of the HMD user's head movements is also required. When he/she moves their head, display lag will generate visual vs non-visual conflicts about their head pose. Palmisano et al. [1] refer to these sensory input conflicts as Differences in Virtual and Physical head pose (or **DVP**). According to their *DVP hypothesis*, sickness should only be induced when user head-movements and display lag generate large amplitude, time-varying patterns of DVP. This prediction has thus

far been supported by the findings of several studies [3, 4]. However, those studies only examined the sickness induced by adding artificial constant display lags. Thus, in the current study, we examined whether the *DVP hypothesis* could also explain HMD user experiences of cybersickness induced by adding time-varying display lags.

2 METHOD

2.1 Participants

33 participants were recruited from the University of Wollongong and general population (20 females and 13 males ranging in age from 18 to 35 years). All were without visual, neurological, or vestibular impairments, and had normal or corrected-to-normal vision.

2.2 Design and Materials

There were 3 blocks of 4 trials. On each trial, participants made continuous 0.5Hz head movements about 1 of 3 cardinal axes (timed to an auditory metronome) while they were exposed to simulations with different types of display lag. Trials had either no added lag (*Baseline* lag), a constant added lag of 222 ms (*Constant* lag) or one of two types of time-varying added lag [the imposed lag in these conditions alternated between 0 and 267 ms every 5.6 s (*0.2Hz Sinusoidal* lag) or every 2.8 s (*0.4Hz Sinusoidal* lag)]. The average effective baseline lag of our PC VR system was ~4 ms when running this custom code using an Oculus Rift CV1 HMD. Sickness on each trial was measured using the Fast Motion Sickness (FMS) scale [5].

2.3 Procedure

On each trial, participants made continuous oscillatory (yaw/pitch/roll) head-movements for 35 s. The display was black for the first 5 s – except for a head-stationary fixation point. Then a virtual room appeared in their HMD and remained visible for the next 30 s. After they had rated the severity of the sickness for that trial, they were given a 3-min rest before the next trial. When they had completed all of the trials for that block, they waited at least 10-min before commencing the next block of trials. This was then repeated for the 3rd block of trials.

2.4 DVP Estimation

HMD sensor and lag data were stored and used to estimate the participant's physical (i.e., current) and virtual (i.e., lagged) 3-D head orientation every 11 ms during each trial. DVP at each of these times was the minimum angle of rotation between the quaternions specifying these 2 corresponding head poses. We next calculated the standard deviation of the DVP (*stdDVP*) to quantify the amount of sensory conflict experienced on each trial. We also performed detrended fluctuation analyses (DFA) on these data [6]. The scaling component (*aDVP*) of each DFA, provided non-redundant information about the temporal dynamics of the DVP for that trial.

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3 RESULTS

3.1 Lag condition effects on cybersickness

A linear mixed model (with random intercepts across participants) was used to examine the relationships between *Lag Condition*, *Head-Movement Axis* and *Sickness Severity* ratings. This revealed a significant effect of *Lag Condition*, $F(3,352) = 96.024$, $p < 0.001$ (Figure 1). Bonferroni-corrected comparisons confirmed that sickness was less severe in *Baseline*, compared to added lag, conditions ($p < 0.001$). However, sickness was similar for *Constant* and *Sinusoidal* added lag conditions ($p \geq 0.702$ for all comparisons). No other main effects or interactions were significant.

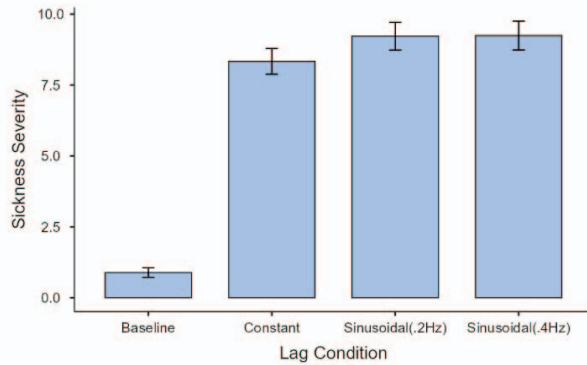


Figure 1: Effects of *Lag Condition* on *Sickness Severity*.

3.2 Predicting sickness severity using DVP

We then examined whether our DVP measures could predict the user's sickness severity (SS). We used the following linear mixed model, which included *stdDVP* and *aDVP* as fixed effects, and had random intercepts and slopes across participants:

$$SS \sim 1 + stdDVP + aDVP + (1 + stdDVP + aDVP | \text{Participant})$$

This model accounted for 46% of the variance in SS ratings – with both *stdDVP* [$F(1,44.9) = 25.4$, $p < 0.001$] and *aDVP* [$F(1,375) = 13.7$, $p < 0.001$] contributing significantly to these predictions. Figure 2 shows that an increase of 1° in *stdDVP* produced an increase of 0.55 in SS ratings. Similarly, Figure 3 shows that an increase of 0.1 in *aDVP* resulted in an increase of 1.7 in SS ratings.

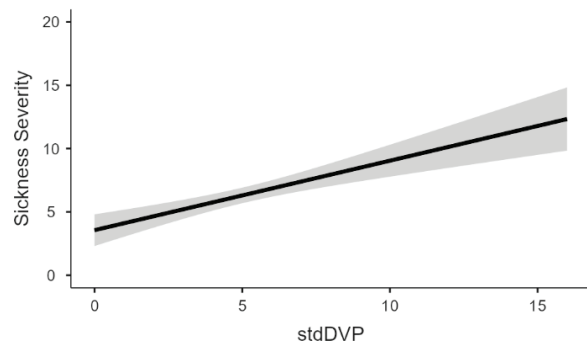


Figure 2: Mean slope (bold line) and 95% CIs (shaded region) for the relationship between *stdDVP* and *Sickness Severity*.

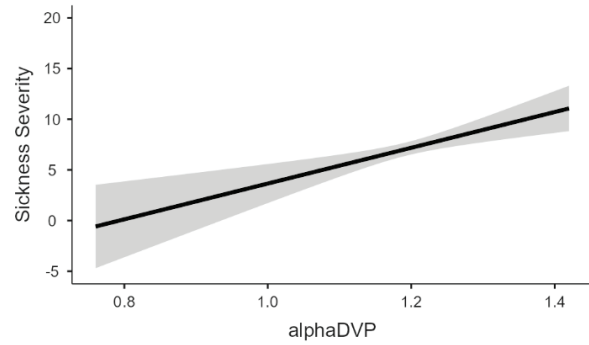


Figure 3: Mean slope (bold line) and 95% CIs (shaded region) for the relationship between *aDVP* and *Sickness Severity*.

4 CONCLUSION

As expected, adding extra display lag to our VR simulations was found to significantly increase cybersickness. However, contrary to St. Pierre et al. [2], sinusoidal lag was not more provocative than constant lag. Indeed, cybersickness was found to be quite similar for all 3 types of added lag in this study.

Studies have previously shown that objective estimates of the DVP can predict the severity of cybersickness produced by adding constant lags to VR simulations. Here we replicated these findings and showed that DVP can also be used to predict the sickness induced by time-varying display lags. Consistent with the *DVP hypothesis*, both *stdDVP* and *aDVP* were found to contribute to predictions of sickness severity – for all 3 types of head-movement examined (i.e., yaw, pitch and roll). The research shows that the spatial magnitude and the temporal dynamics of the DVP both provide useful information for predicting cybersickness.

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REFERENCES

- [1] S. Palmisano, R.S. Allison, and J. Kim. Cybersickness in Head-Mounted Displays is caused by differences in the user's virtual and physical head pose. *Frontiers in Virtual Reality*, 1:587698, 2020. doi: 10.3389/frvir.2020.587698.
- [2] M.E. St. Pierre, S. Banerjee, A.W. Hoover, and E.R. Muth. The effects of 0.2 Hz varying latency with 20–100 ms varying amplitude on simulator sickness in a helmet mounted display. *Displays*, 36:1–8, 2015. doi: 10.1016/j.displa.2014.10.005.
- [3] J. Kim, W. Luu, and S. Palmisano. Multisensory integration and the experience of scene instability, presence and cybersickness in virtual environments. *Computers in Human Behavior*, 113:106484, 2020. doi: 10.1016/j.chb.2020.106484.
- [4] S. Palmisano, R.S. Allison, J. Teixeira, and J. Kim. Differences in virtual and physical head orientation predict sickness during active head-mounted display based virtual reality. *Virtual Reality*, 27:1293–1313, 2023. doi: 10.1007/s10055-022-00732-5.
- [5] B. Keshavarz, and H. Hecht. Validating an efficient method to quantify motion sickness. *Human Factors*, 53:415–426, 2011. doi: 10.1177/0018720811403736.
- [6] C.K. Peng, S. Havlin, H.E. Stanley, and A.L. Goldberger. Quantification of scaling exponents and crossover phenomena in nonstationary heartbeat time series. *Chaos*, 5(1):82–87, 1995. doi: 10.1063/1.166141.